

Deepti Singh

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EDUCATION

Thakur College of Engineering and Technology
Bachelor of Technology in Artificial Intelligence and Data Science

Mumbai, Maharashtra
Nov. 2022– May 2026

Bharatiya Vidya Bhavans
Science

Mumbai, Maharashtra
Aug. 2020 – June 2022

EXPERIENCE

AI Engineer - Intern

Vedic Rishi Astro

August 2025 – May 2026

Mumbai, Maharashtra

- Independently researched and built a 4-agent CrewAI pipeline from scratch powered by GPT-4 with specialized agents for data retrieval, astrological reasoning, PDF report design, and end-to-end validation, automating the full report generation workflow for the content team.
- Reduced manual content team effort by 30% by replacing fully manual reporting processes with LLM-based multi-agent orchestration enabling non-technical teams to focus on high-value creative tasks.
- Trained a YOLOv8 object detection model on a 7,100-image dataset over 75 epochs to detect and classify 4 palm line features (heart, head, life, and fate lines) achieving 82% accuracy and handling real-world challenges like partial and occluded lines.
- Deployed both the CrewAI pipeline and YOLOv8 model as containerized services on AWS Lambda using Docker enabling scalable, cost-effective serverless execution in production.
- Designed data flow across all agents using PostgreSQL for persistent storage and Redis for caching ensuring consistent state management and reliable AI model integration throughout the pipeline.

Backend Developer

GNZY

November 2024 – December 2024

Mumbai, Maharashtra

- Streamlined backend codebase, reducing technical debt by 25% and improving system performance by 15%.
- Designed and tested 20+ APIs, achieving 100percent functionality and data accuracy within project deadlines.

PROJECTS

Palm Line Detection System | YOLOv8, AWS Lambda, Docker

Jan 2026

- Built a model using YOLOv8 dataset to detect and classify 4 palm line features (heart, head, life and fate lines) trained on a 7,100 image dataset over 75 epochs achieving 82% accuracy.
- Handled real-world challenges including partial, occluded and faint palm lines across diverse hand types iterated across multiple training runs (100 epoch vs 75 epoch) to find optimal performance.
- Deployed the trained model on AWS Lambda using Docker as a serverless inference endpoint enabling real-time palm line detection in production.

TrustED: AI-Based Accreditation Sytem | Python, Pandas, CrewAI

Jan 2025-Present

- Building a desktop app using Python and Tkinter to help colleges evaluate NBA Self-Assessment Reports.
- Automates data validation and report generation, aiming to cut manual work by 50–60%.
- Migrating from Tkinter to CrewAI-based multi-agent architecture for significantly improved automation and scalability.

TECHNICAL SKILLS

Languages: Python, JavaScript, Typescript, HTML/CSS

Frameworks: React.js, Node.js, Express.js, FastAPI, Flask, Streamlit, Next.js, CrewAI

Developer Tools: GitHub, Jupyter Notebook, PostgreSQL, MySQL, MongoDB, Postman, Power BI, Tableau, Redis, AWS Lambda, Docker

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Beautiful Soup, Requests

AI for Detecting Greenwashing in Corporate Sustainability Reports

Accepted

Research Project

Jan 2025-April 2025

- Conceptualized and designed a web-based system utilizing NLP models to identify misleading corporate sustainability claims.
- Defined workflows, UI mockups, backend logic, and data architecture for features like claim verification, scorecard exploration, and community reporting.
- Aimed to reduce manual greenwashing detection by up to 70%, enhancing transparency and enabling faster assessments.
- The research is under the guidance of Dr.Anand Khandare(Associate Dean of Planning and Operations for Digital Resources, TCET).

TrustED: AI-Powered Accreditation System

In-Process

Research Project

Jan 2025 - Present

- Developing **TrustED**, an AI-driven system that streamlines accreditation processes in higher education.
- Uses NLP and machine learning to evaluate Self-Assessment Reports and provide accreditation readiness scores and insights.
- Enhances efficiency, reduces manual effort, and offers a transparent alternative to traditional accreditation methods.
- The research is under the guidance of Dr.Anand Khandare(Associate Dean of Planning and Operations for Digital Resources, TCET).